JUNHOO LEE

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RESEARCH EXPERIENCE

KAIST Sep 2026 – Present
Postdoctoral Researcher
Working with Prof. Jinwoo Shin.

EDUCATION

Seoul National University Seoul, South Korea
Ph.D. in Intelligence and Information Sep 2021 – Aug 2026
Advisor: Prof. Nojun Kwak, Machine Intelligence & Pattern Analysis Laboratory (MIPAL)
Dissertation: *Projection, Persistence, and Elicitation of Learned Internal Structure*
Best Ph.D. Dissertation Award, GSCT representative awardee (1 of 27 Ph.D. graduates).

Seoul National University Seoul, South Korea
B.Sc. in Electrical and Computer Engineering Mar 2017 – Sep 2021
B.S. Thesis: *Improvement of Elevator Control Algorithm Using Reinforcement Learning*. Advisor: Sungroh Yoon

RESEARCH INTERESTS

I view AI through the lens of self-reference. Many of its remarkable capabilities emerge from recursive structures in which a system's own outputs, representations, or experiences feed back into its subsequent operation. I am fascinated by why these structures appear across different levels of AI, how they give rise to new capabilities, and what makes them work.
My current research focuses on robot harnesses, looped models, and continual harnessing, while I remain open to exploring other directions.

PUBLICATIONS

*: Equal contribution.

[C1] **Discriminative Score Function: Turning Pretrained Models into Functional Generative Priors**
Junhoo Lee, *Hyeonjin Kim, Sangbum Han, Nojun Kwak*
The Conference on Neural Information Processing Systems (NeurIPS), 2026.

[C2] **Clean Data Can Still Carry Backdoors: Support-Persistent Backdoors for Model Reuse**
Junhoo Lee*, *Baekseung Kim**, *Seungyeon Kim, Nojun Kwak*
The Conference on Neural Information Processing Systems (NeurIPS), 2026.

[C3] **ManiFusion: Unlocking High-Throughput Generation via Superposition in Manifold Space**
Minkyu Kim, Baekseung Kim, **Junhoo Lee**, *Jangho Kim, Nojun Kwak*
The Conference on Neural Information Processing Systems (NeurIPS), 2026.

[C4] **Unlocking the Potential of Diffusion Language Models through Template Infilling**
Junhoo Lee, *Seungyeon Kim, Nojun Kwak*
Annual Meeting of the Association for Computational Linguistics (ACL), 2026. **Oral Presentation.**

[C5] **CSF: Black-box Fingerprinting via Compositional Semantics for Text-to-Image Models**
Junhoo Lee, *Mijin Koo, Nojun Kwak*
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

[C6] **Deep Edge Filter: Return of the Human-Crafted Layer in Deep Learning**
*Dongkwan Lee**, **Junhoo Lee***, *Nojun Kwak*
The Conference on Neural Information Processing Systems (NeurIPS), 2025.

[C7] **What's Making That Sound Right Now? Video-centric Audio-Visual Localization**
Hahyeon Choi, **Junhoo Lee**, *Nojun Kwak*
The IEEE/CVF International Conference on Computer Vision (ICCV), 2025.

[C8] **Deep Support Vectors**
Junhoo Lee, *Hyunho Lee, Kyomin Hwang, Nojun Kwak*
The Conference on Neural Information Processing Systems (NeurIPS), 2024.

[C9] **Any-Way Meta Learning**
Junhoo Lee, *Yearim Kim, Hyunho Lee, Nojun Kwak*
The AAAI Conference on Artificial Intelligence (AAAI), 2024.

[C10] **SHOT: Suppressing the Hessian along the Optimization Trajectory for Gradient-Based Meta-Learning**
Junhoo Lee, *Jayeon Yoo, Nojun Kwak*
The Conference on Neural Information Processing Systems (NeurIPS), 2023.

[J1] **End-to-End Multi-Entity Customization**
Wonhark Park, Jaehyun Lee*, Wonsik Shin,* **Junhoo Lee**, *Nojun Kwak*
IET Image Processing, 20(1), e70306, 2026.

[J2] **The Role of Teacher Calibration in Knowledge Distillation**
Suyoung Kim, Seonguk Park, **Junhoo Lee**, *Nojun Kwak*
IEEE Access, 13, 115548–115557, 2025.

[W1] **3DROID: A Renderable 3D Gaussian Dataset with Measured Per-Scene Reliability**
Wonguen Cho, **Junhoo Lee**, *Nojun Kwak*
The 1st Workshop on Physical World AI: Geometry, Characteristics, and Multimodal Sensing at the Conference on Neural Information Processing Systems (NeurIPS), 2026.

[W2] **Completion Aware Guidance for World Action Models**
Seungyeon Kim, **Junhoo Lee**, *Baekseung Kim, Minkyu Kim, Nojun Kwak*
Robotics World Modeling Workshop at the Conference on Neural Information Processing Systems (NeurIPS), 2026.

[W3] **Point-to-Point: Sparse Motion Guidance for Controllable Video Editing**
Yeji Song, Jaehyun Lee, Mijin Koo, **Junhoo Lee**, *Nojun Kwak*
The 4th AI for Visual Arts Workshop at the European Conference on Computer Vision (AI4VA, ECCV Workshops), 2026.

[W4] **Do Not Think About Pink Elephant!**
Kyomin Hwang, Suyoung Kim*,* **Junhoo Lee***, *Nojun Kwak*
The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR Workshops, Responsible Generative AI), 2024.

[W5] **Coreset Selection for Object Detection**
Hojun Lee, Suyoung Kim, **Junhoo Lee**, *Jaeyoung Yoo, Nojun Kwak*
The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPR Workshops, Dataset Distillation), 2024.

[W6] **Practical Dataset Distillation Based on Deep Support Vectors**
Hyunho Lee, **Junhoo Lee**, *Nojun Kwak*
The European Conference on Computer Vision Workshops (ECCV Workshops, Dataset Distillation Challenge), 2024.

[P1] **RobotUse: Allocating Computation, Context, and Decisions**
Junhoo Lee, *Injun Baek, Seungyeon Kim, Suhyun Jeon, Minkyu Kim, Baekseung Kim, Nojun Kwak*
arXiv preprint arXiv:2610.04929, 2026.

[P2] **Recursive Harness Distillation across Agents for Robot Manipulation**
Seungyeon Kim, **Junhoo Lee**, *Minkyu Kim, Baekseung Kim, Nojun Kwak*
arXiv preprint arXiv:2609.33378, 2026.

[P3] **ACRO: Actor–Critic Rollout Orchestration**
Junhoo Lee, *Seungyeon Kim, Baekseung Kim, Minkyu Kim, Suhyun Jeon, Jimyeong Kim, Nojun Kwak*
Preprint, 2026.

AWARDS & HONORS

Best Ph.D. Dissertation Award, GSCT, Seoul National University	2026
InnoCore Fellowship	2026
NeurIPS Top Reviewer	2026
ICML Gold Reviewer	2026
BK21 Future Innovation Talent Bronze Prize (USD 1,000)	2023
BK21 Outstanding Research Talent Fellowship (USD 3,500)	2023
Yulchon AI Star Scholarship (USD 8,000)	2022
3rd Place, SNU FastMRI Challenge (USD 3,000)	2021
Kwanak Scholarship (full funded)	2021
National Science and Engineering Scholarship (full funded)	2017–2018

RESEARCH PROJECTS

Gradient-Descent-Based Meta-Learning Algorithms (Samsung Electronics, Main Researcher, 2022–2023)
Self-Directed Visual Object Detection for Shopping Assistance (IITP, Project Manager, 2022–2024)
AI Video Generation for Korean Traditional Media Art (MCST/EASYWITH, Main Engineer, 2024–2026)
Physically Plausible Video Diffusion Modeling (AI Star Fellowship/MSIT, Project Manager, 2025).

SKILLS

Research Development: Python, PyTorch, CUDA; **Experiment Infrastructure**: Slurm, Docker.

TALKS

Compositional Semantic Fingerprinting (CSF) for Black-box Attribution of Text-to-Image Models, SWCS 2026, Apr. 2026
Deep Support Vectors: Interpreting Deep Models via Support Vector Extraction, KCC 2024, Jun. 26, 2024.

ACADEMIC SERVICE

Top Reviewer: NeurIPS 2026.
Gold Reviewer: ICML 2026.
Conference Reviewer: CVPR (2025, 2026), ICCV (2025), ECCV (2026), ICLR (2025, 2026), ICML (2026), NeurIPS (2024, 2025, 2026).

PATENTS

Method and Apparatus for Edge Filtering to Improve the Robustness of Deep Learning Models, Korean Patent 10-2025-0215458
Method and Apparatus for Controlling Unintended Object Generation in Image Generation Models through Prompt Reconfiguration, Korean Patent 10-2025-0063788.